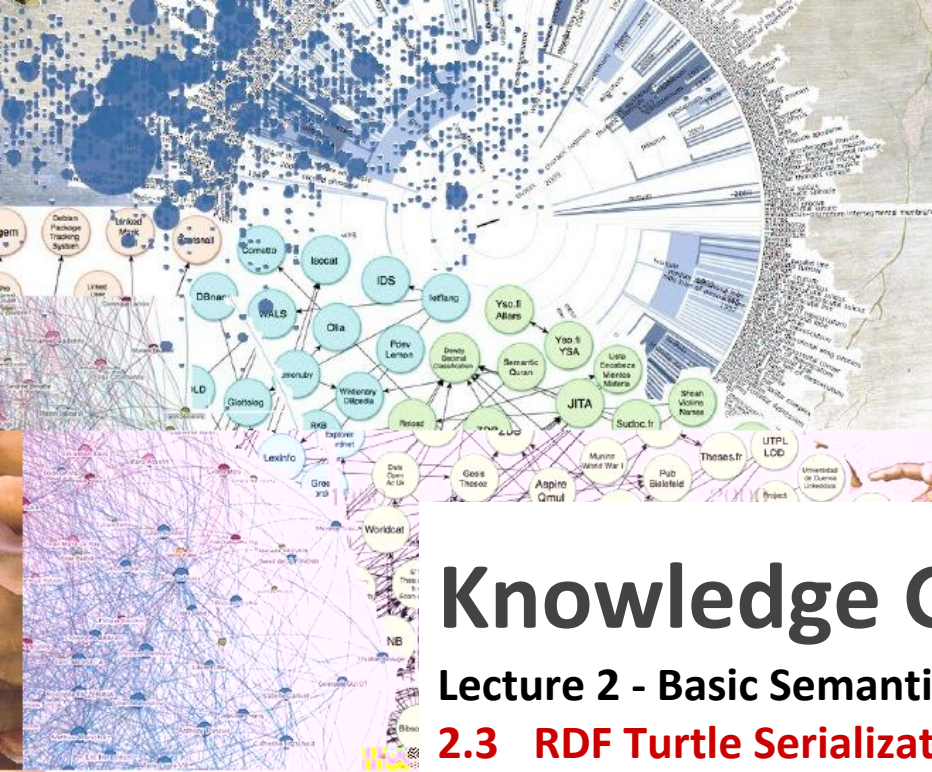




Knowledge Graphs

Lecture 2 - Basic Semantic Technologies

2.3 RDF Turtle Serialization

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FIZ Karlsruhe - Leibniz Institute for Information Infrastructure

AIFB - Karlsruhe Institute of Technology

Autumn 2020



FIZ Karlsruhe

Leibniz-Institut für Informationsinfrastruktur

Knowledge Graphs

Lecture 2: Basic Semantic Technologies

2.1 How to Identify and Access Things

2.2 How to Represent Simple Facts with RDF

2.3 RDF Turtle Serialization

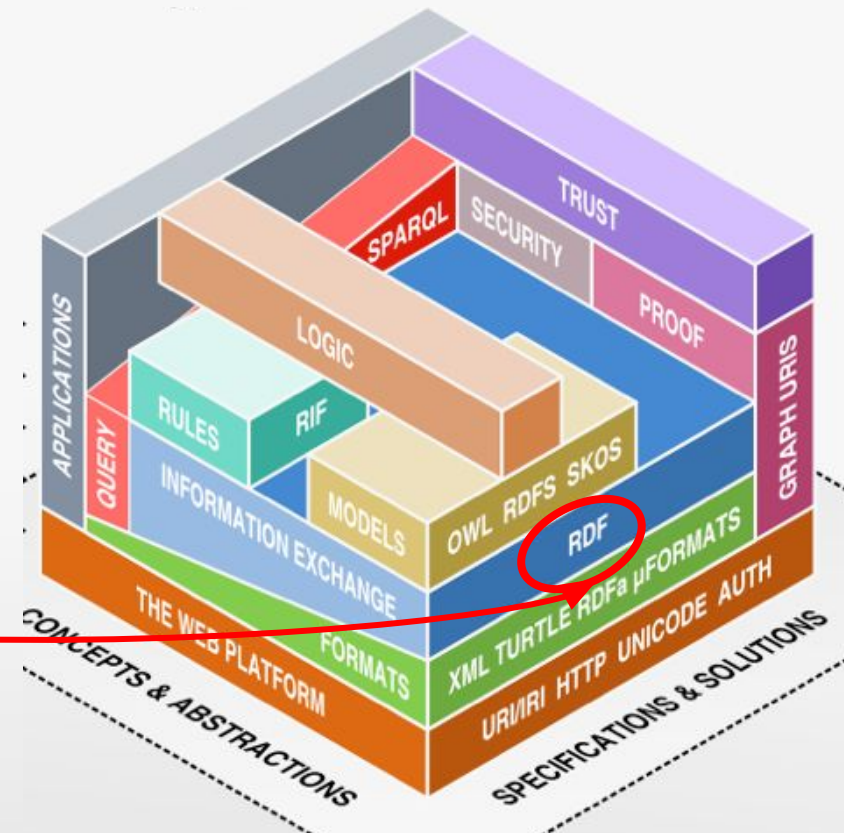
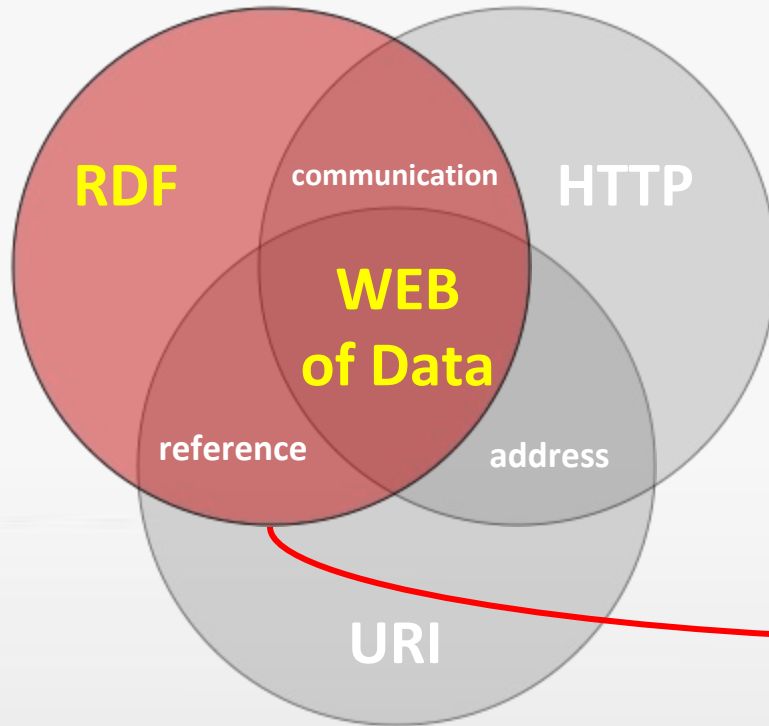
2.4 RDF Complex Data Structures

2.5 Model Building with RDFS

2.6 Logical Inference with RDF(S)

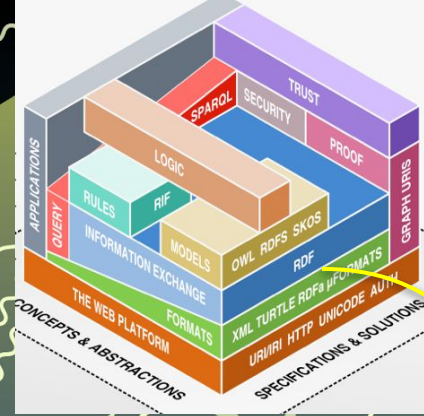
Excursion 1: RDFa - RDF and the Web

Basic Architecture of the Web of Data



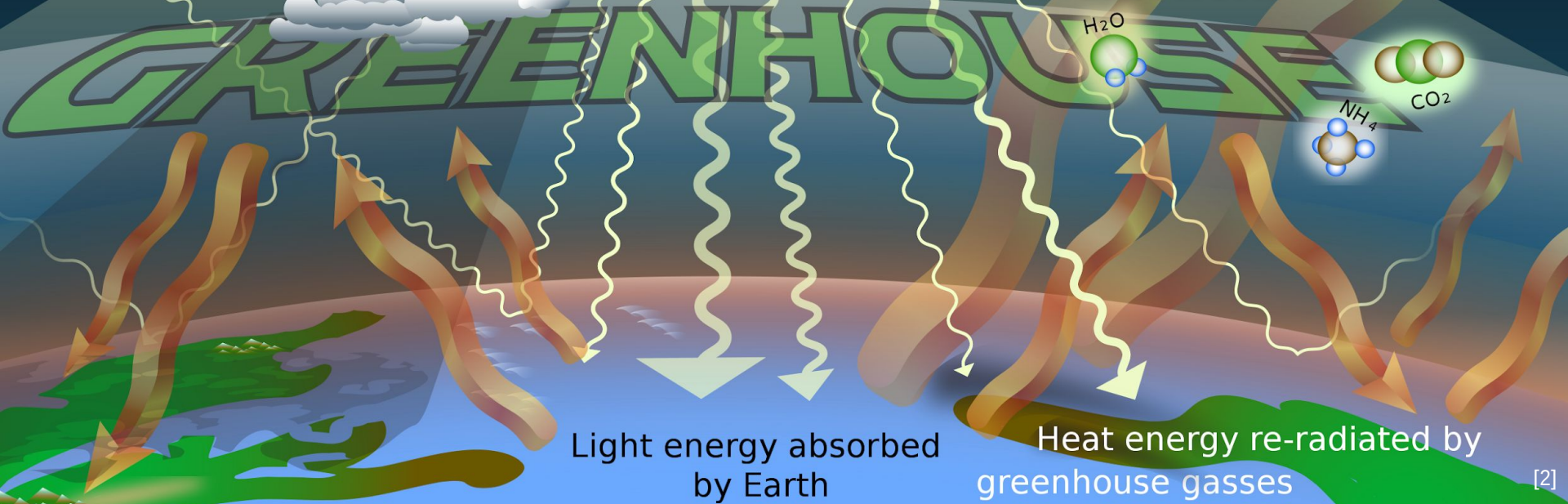
Heat energy radiated into space

Resource Description Framework



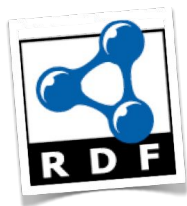
Reflected from surface

Reflected from cloudcover



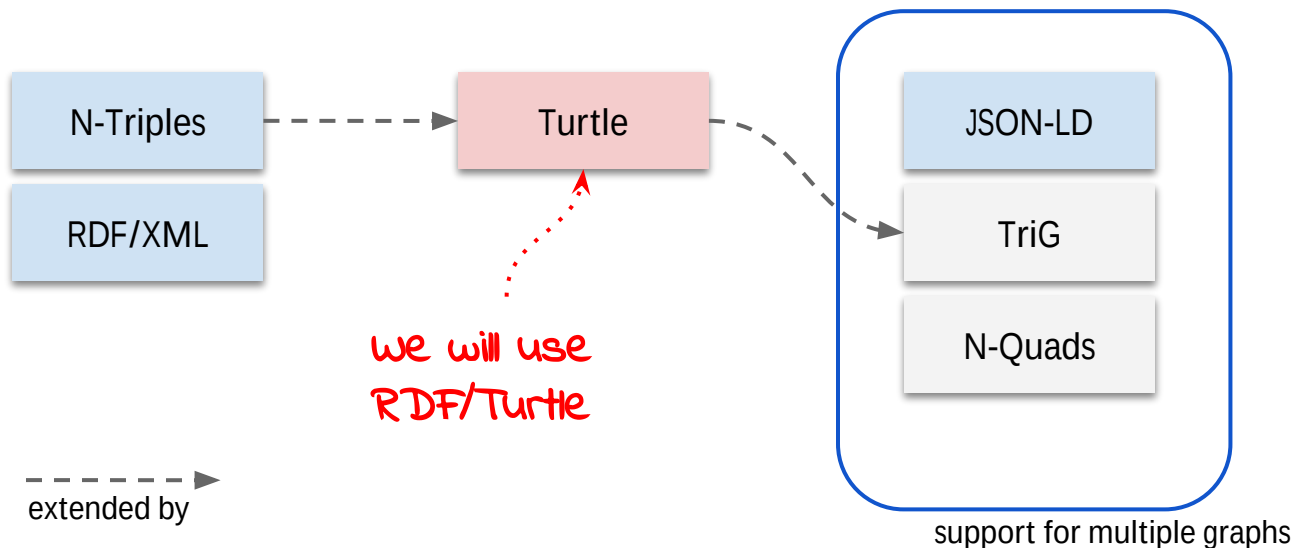
Light energy absorbed by Earth

Heat energy re-radiated by greenhouse gasses



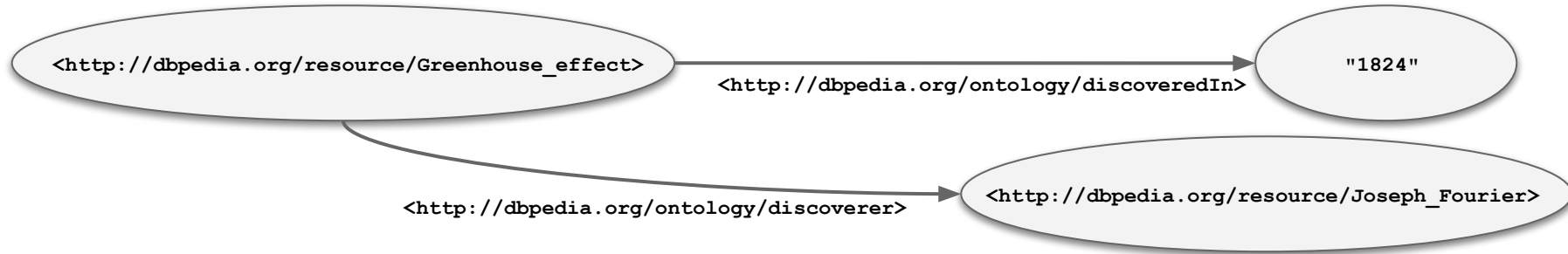
RDF Serializations

- RDF comes with several different **serialization formats**:
 - N-Triples, RDF/XML, JSON, Turtle, TriG, N-Quads, RDFa, ...





RDF Serializations



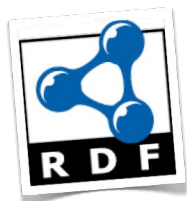
- **N-Triples Notation**

- **URIs/IRIs** in angle brackets
- **Literals** in quotation marks
- Triple ends with a **period**

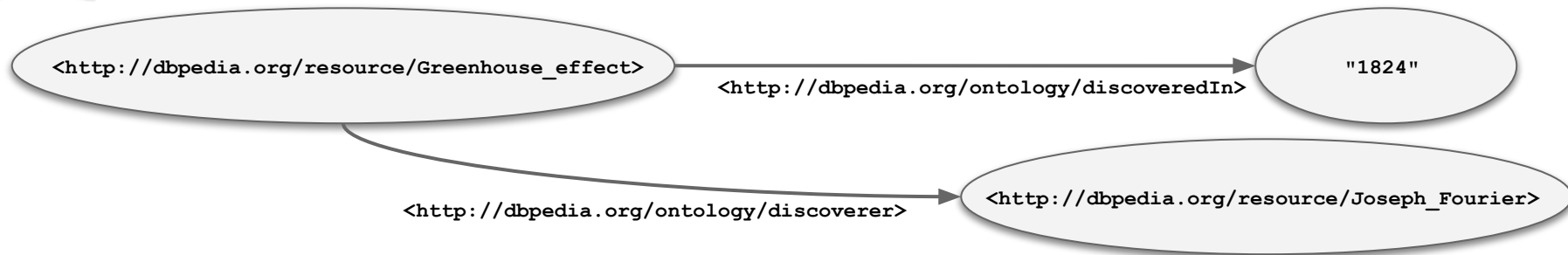
```
<http://dbpedia.org/resource/Greenhouse_effect> <http://dbpedia.org/ontology/discoveredIn> "1824" .
```

```
<http://dbpedia.org/resource/Greenhouse_effect> <http://dbpedia.org/ontology/discoverer>
```

```
<http://dbpedia.org/resource/Joseph_Fourier> .
```



RDF Serializations



- **Turtle (Terse RDF Triple Language) Notation**
 - extension of N-Triples

```
@prefix dbo: <http://dbpedia.org/ontology/> .
@base <http://dbpedia.org/resource/> .

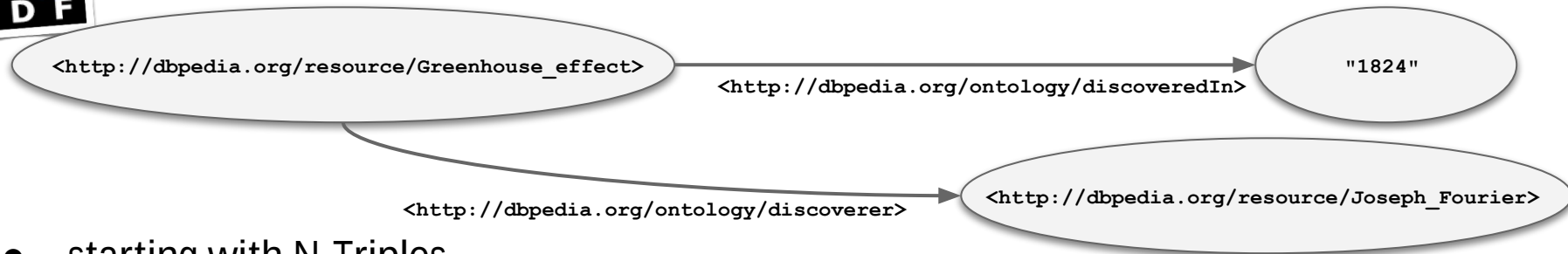
<Greenhouse_effect> dbo:discoveredIn "1824" .

<Greenhouse_effect> dbo:discoverer <Joseph_Fourier> .
```

RDF/Turtle allows
shortcuts and
abbreviations for
readability



RDF/Turtle



starting with N-Triples

```
<http://dbpedia.org/resource/Greenhouse_effect> <http://dbpedia.org/ontology/discoveredIn> "1824" .
<http://dbpedia.org/resource/Greenhouse_effect> <http://dbpedia.org/ontology/discoverer>
<http://dbpedia.org/resource/Joseph_Fourier> .
```

```
@prefix dbo: <http://dbpedia.org/ontology/> .
@base <http://dbpedia.org/resource/> .

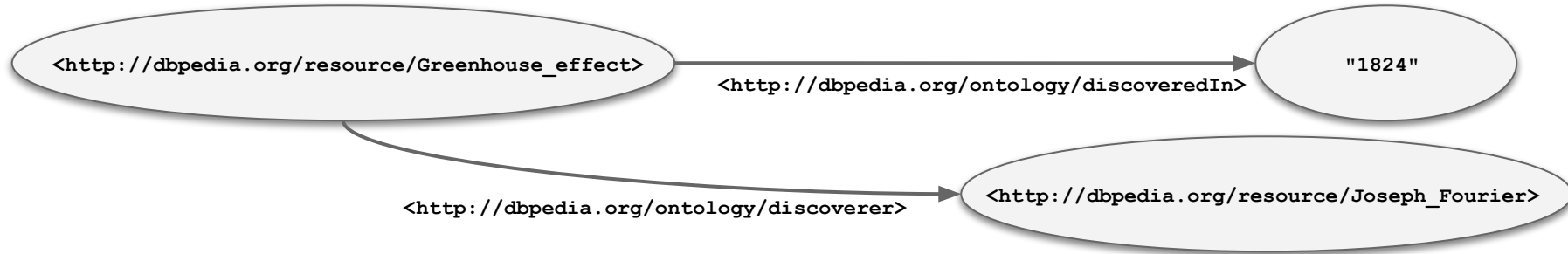
<Greenhouse_effect> dbo:discoveredIn "1824" .
<Greenhouse_effect> dbo:discoverer <Joseph_Fourier> .
```

@prefix directive associates prefix-label with URI

@base directive provides URI to complement all relative URIs



RDF/Turtle

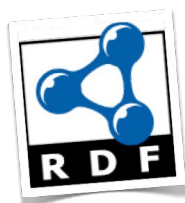


- Abbreviating RDF Triples in Turtle

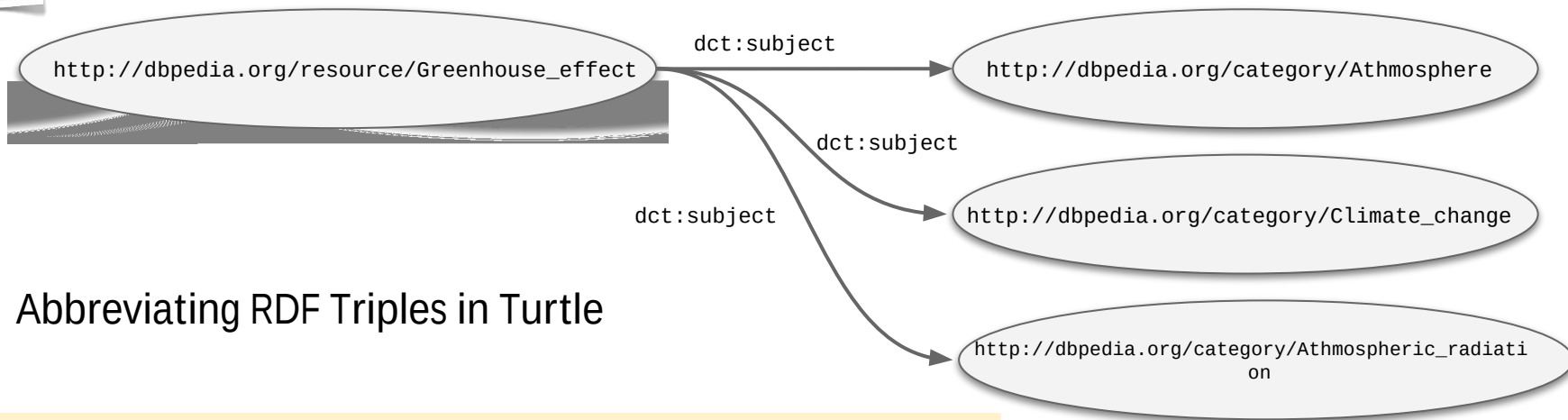
```
@prefix dbo: <http://dbpedia.org/ontology/> .
@base <http://dbpedia.org/resource/> .

<Greenhouse_effect> dbo:discoveredIn "1824" ;
                    dbo:discoverer <Joseph_Fourier> .
```

semicolon indicates that subsequent
triples have the same subject
(predicate list)



RDF/Turtle

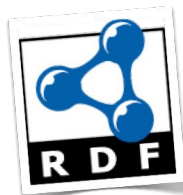


- Abbreviating RDF Triples in Turtle

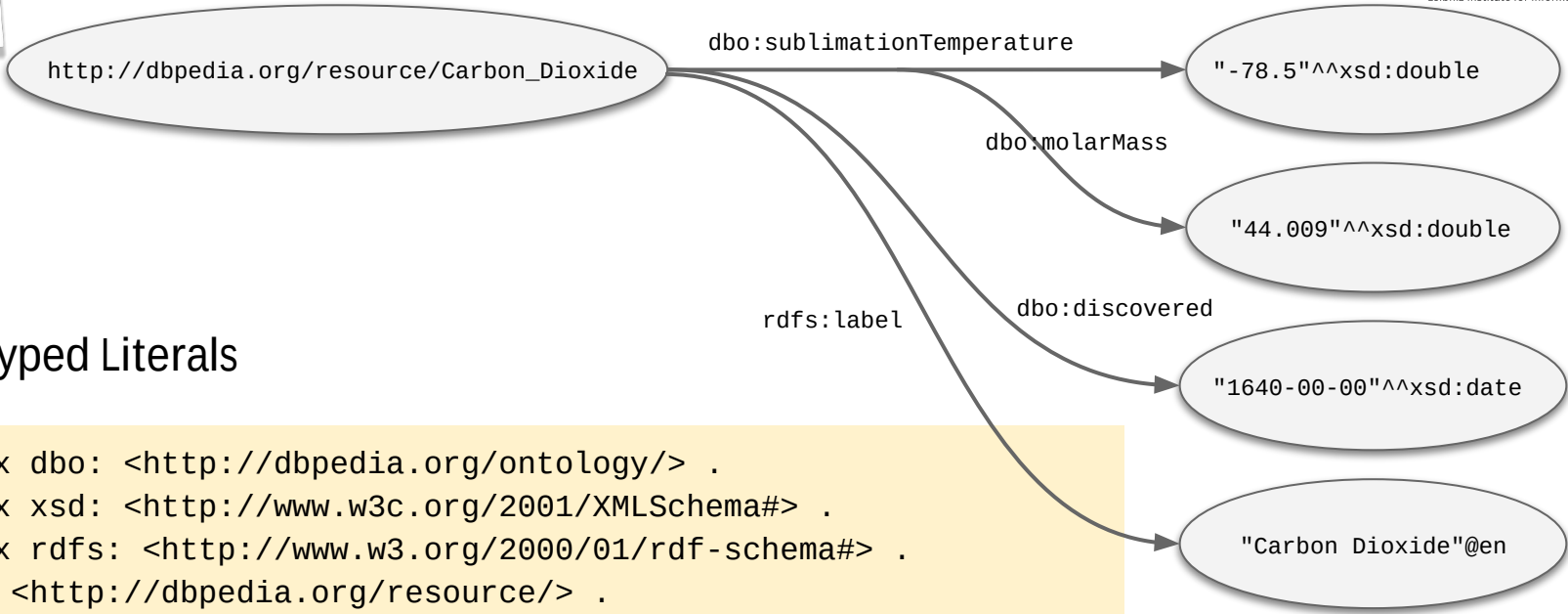
```
@prefix dct: <http://purl.org/dc/terms/> .
@prefix dbc: <http://dbpedia.org/category/> .
@base      <http://dbpedia.org/resource/> .

<Greenhouse_effect> dct:subject dbc:Athmosphere ,
                                dbc:Climate_change ,
                                dbc:Athmospheric_radiation .
```

comma indicates that subsequent triples have same subject and property (object list)



RDF/Turtle



- Typed Literals

```
@prefix dbo: <http://dbpedia.org/ontology/> .
@prefix xsd: <http://www.w3c.org/2001/XMLSchema#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@base <http://dbpedia.org/resource/> .

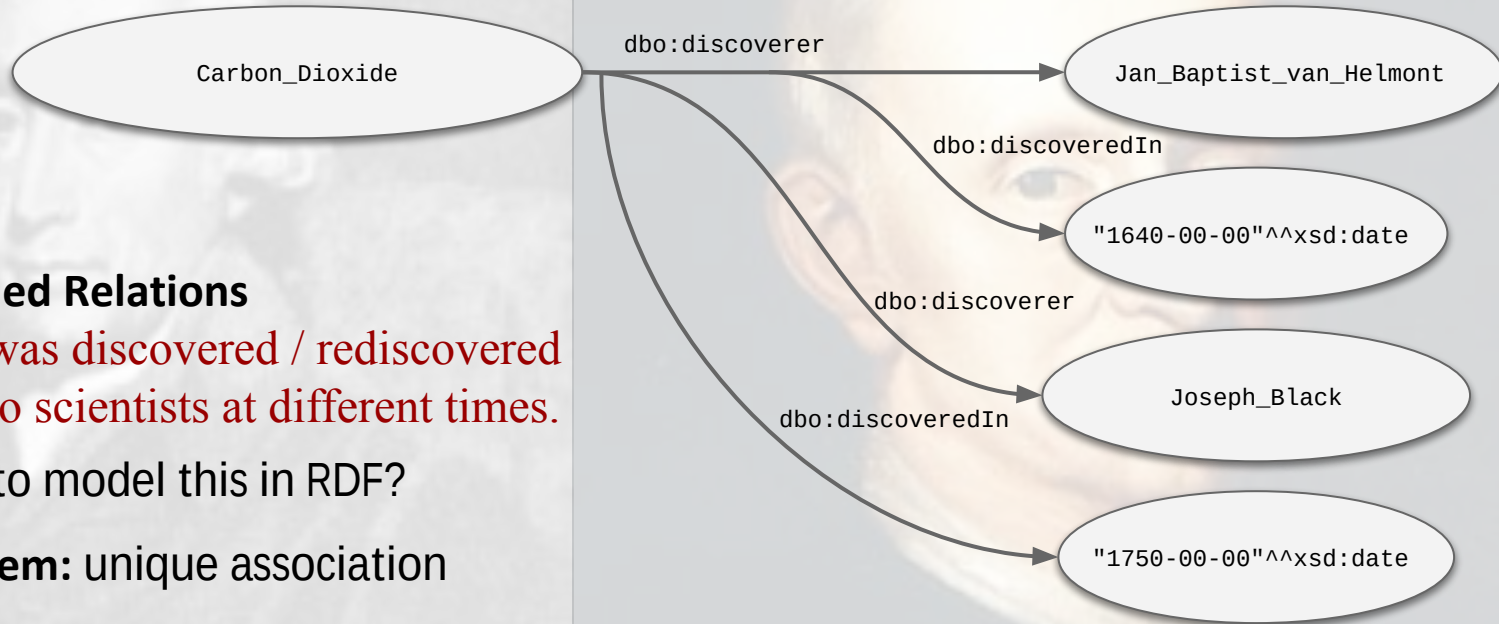
<Carbon_Dioxide> dbo:sublimationTemperature "-78.5"^^xsd:double ;
                  dbo:molarMass "44.009"^^xsd:double ;
                  dbo:discovered "1640-00-00"^^xsd:date ;
                  rdfs:label "Carbon Dioxide"@en .
```



RDF/Turtle

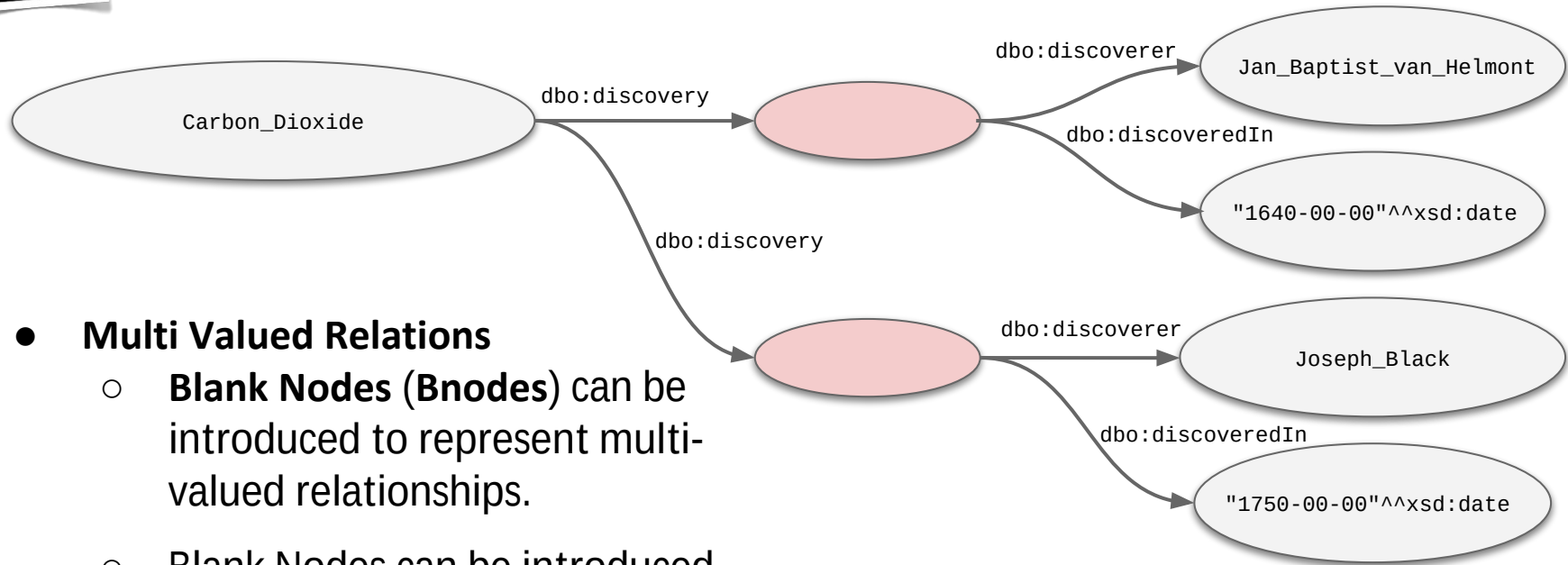
- **Multi Valued Relations**

- CO₂ was discovered / rediscovered by two scientists at different times.
- How to model this in RDF?
- **Problem:** unique association



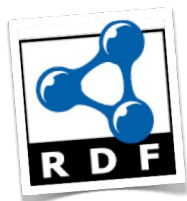


RDF/Turtle

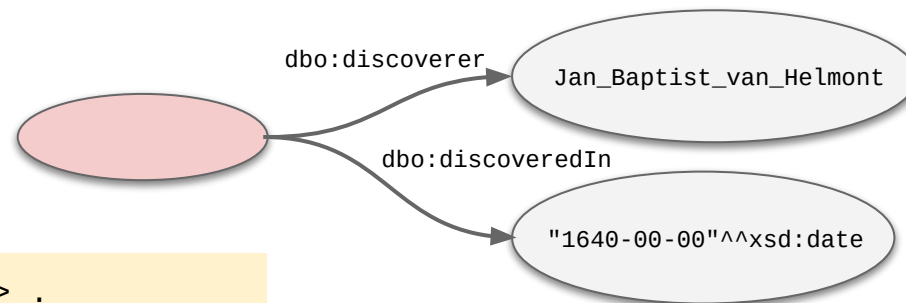


- **Multi Valued Relations**

- **Blank Nodes (Bnodes)** can be introduced to represent multi-valued relationships.
- Blank Nodes can be introduced for resources that don't need a name (auxiliary nodes).



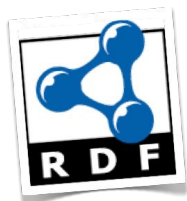
RDF/Turtle



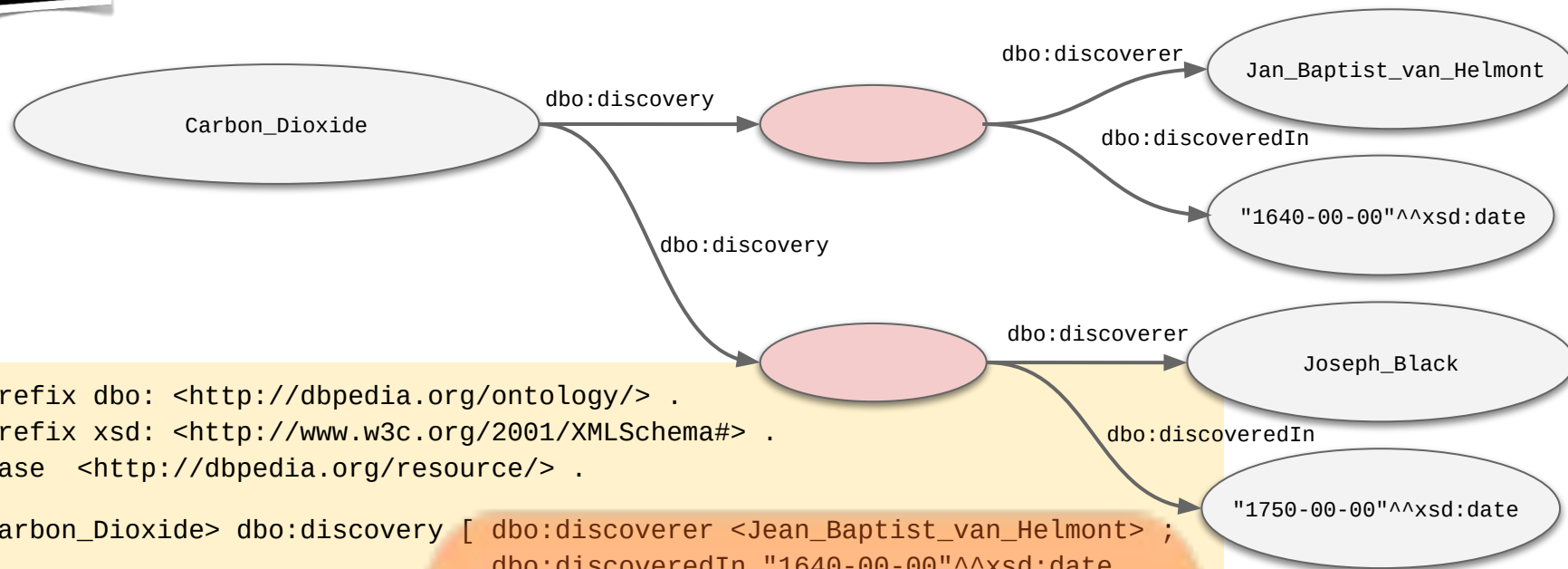
```
@prefix dbo: <http://dbpedia.org/ontology/> .  
@prefix xsd: <http://www.w3c.org/2001/XMLSchema#> .  
@base <http://dbpedia.org/resource/> .
```

```
[ ] dbo:discoverer <Jan_Baptist_van_Helmont> ;  
    dbo:discoveredIn "1640-00-00"^^xsd:date .
```

anonymous blank node as subject



RDF/Turtle



```

@prefix dbo: <http://dbpedia.org/ontology/> .
@prefix xsd: <http://www.w3c.org/2001/XMLSchema#> .
@base <http://dbpedia.org/resource/> .

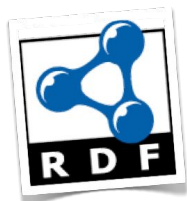
```

```

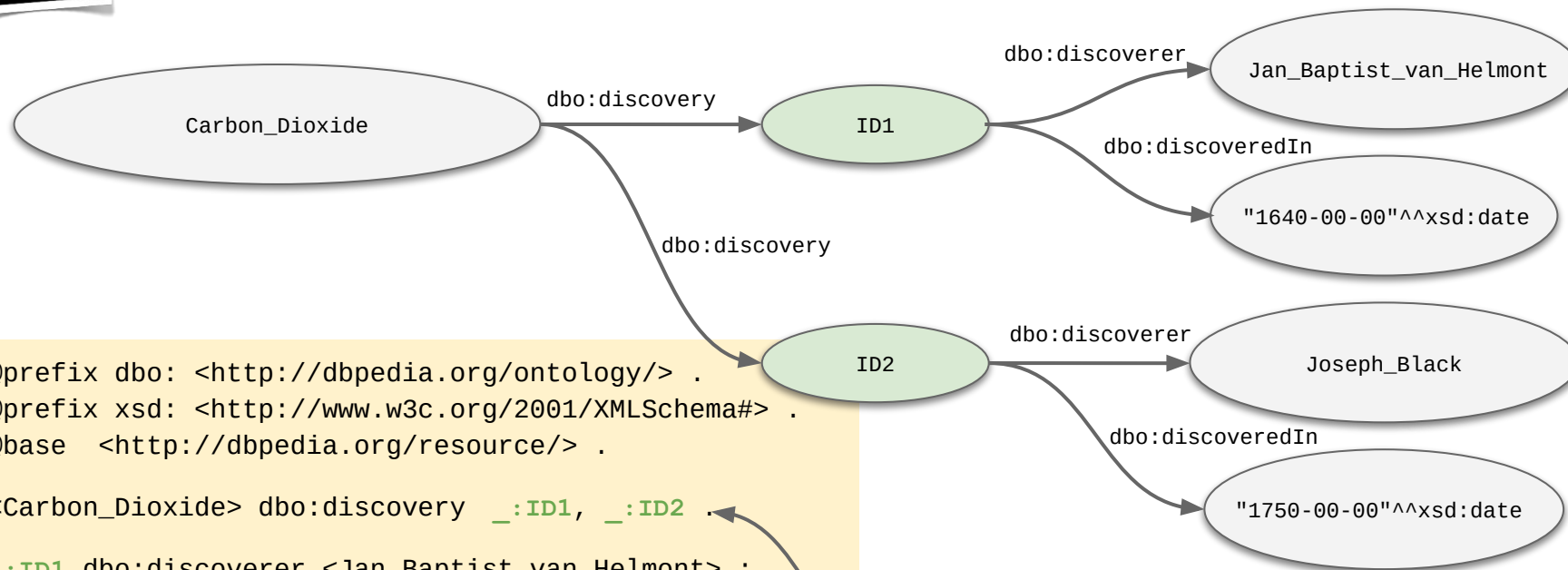
<Carbon_Dioxide> dbo:discovery [
  dbo:discoverer <Jean_Baptist_van_Helmont> ;
  dbo:discoveredIn "1640-00-00"^^xsd:date
] ,
[
  dbo:discoverer <Joseph_Black> ;
  dbo:discoveredIn "1750-00-00"^^xsd:date
] .

```

nested
anonymous blank
nodes



RDF/Turtle



```

@prefix dbo: <http://dbpedia.org/ontology/> .
@prefix xsd: <http://www.w3c.org/2001/XMLSchema#> .
@base <http://dbpedia.org/resource/> .

```

```

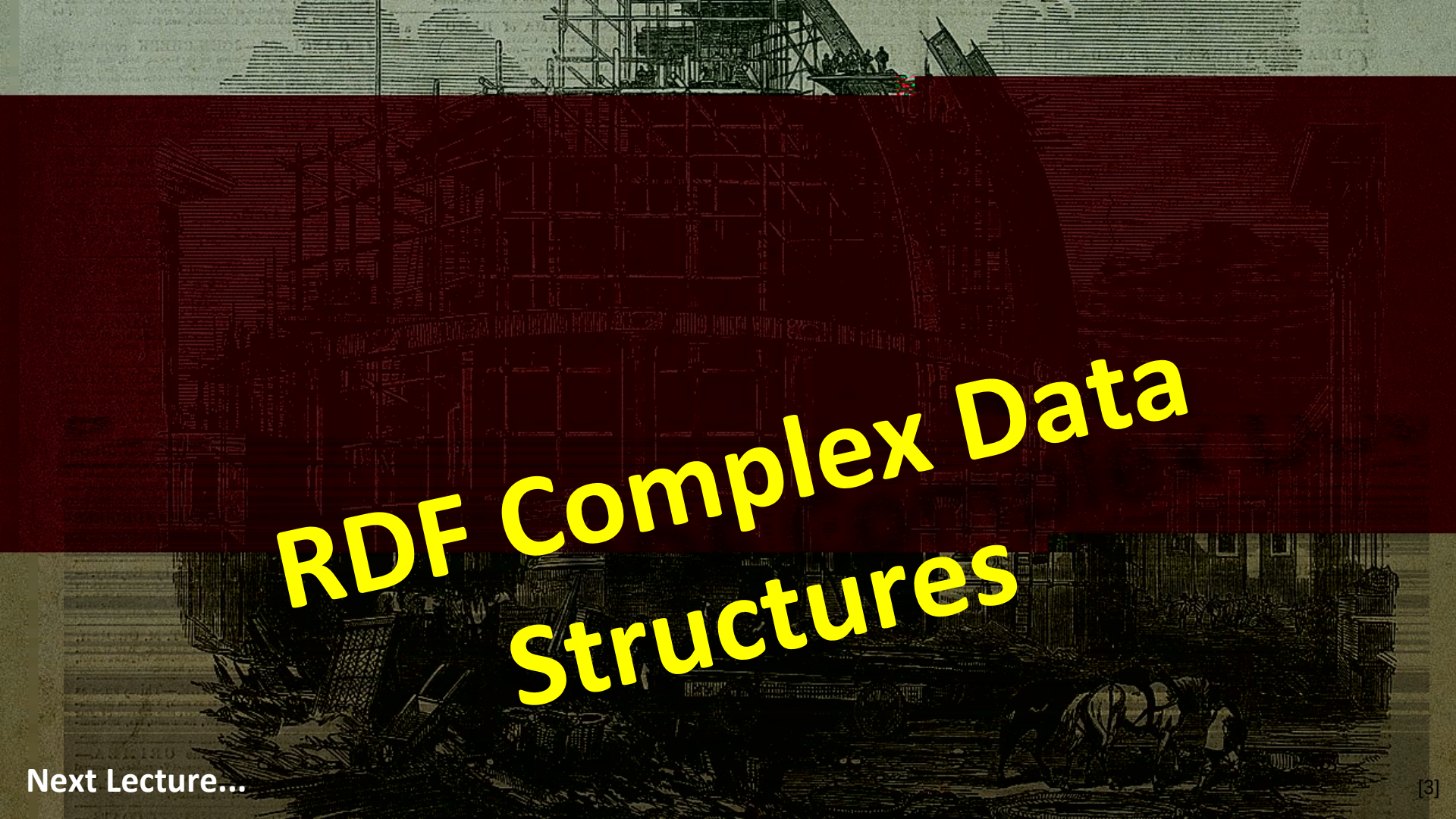
<Carbon_Dioxide> dbo:discovery _:ID1, _:ID2 .

_:ID1 dbo:discoverer <Jan_Baptist_van_Helmont> ;
      dbo:discoveredIn "1640-00-00"^^xsd:date .

_:ID2 dbo:discoverer <John_Black> ;
      dbo:discoveredIn "1750-00-00"^^xsd:date .

```

dereferenceable blank nodes
can only be referenced from
inside a document / graph



RDF Complex Data Structures

Next Lecture...

Picture References:

- [1] Benjamin Nowack, *The Semantic Web - Not a Piece of cake...*, at bnode.org, 2009-07-08 , [CC BY 3.0]
<http://bnode.org/blog/2009/07/08/the-semantic-web-not-a-piece-of-cake>
- [2] The Green House Effect, A loose necktie [CC BY-SA]
<https://commons.wikimedia.org/wiki/File:Greenhouse-effect-t445.svg>
- [3] The British Museum: the reading room under construction. Wood engraving by J. Brown after C. W. Sheeres, 1855., The Wellcome Collection, [CC-BY-4.0]
https://upload.wikimedia.org/wikipedia/commons/a/ad/The_British_Museum%3B_the_reading_room_under_construction._Woo_Wellcome_V0013519.jpg